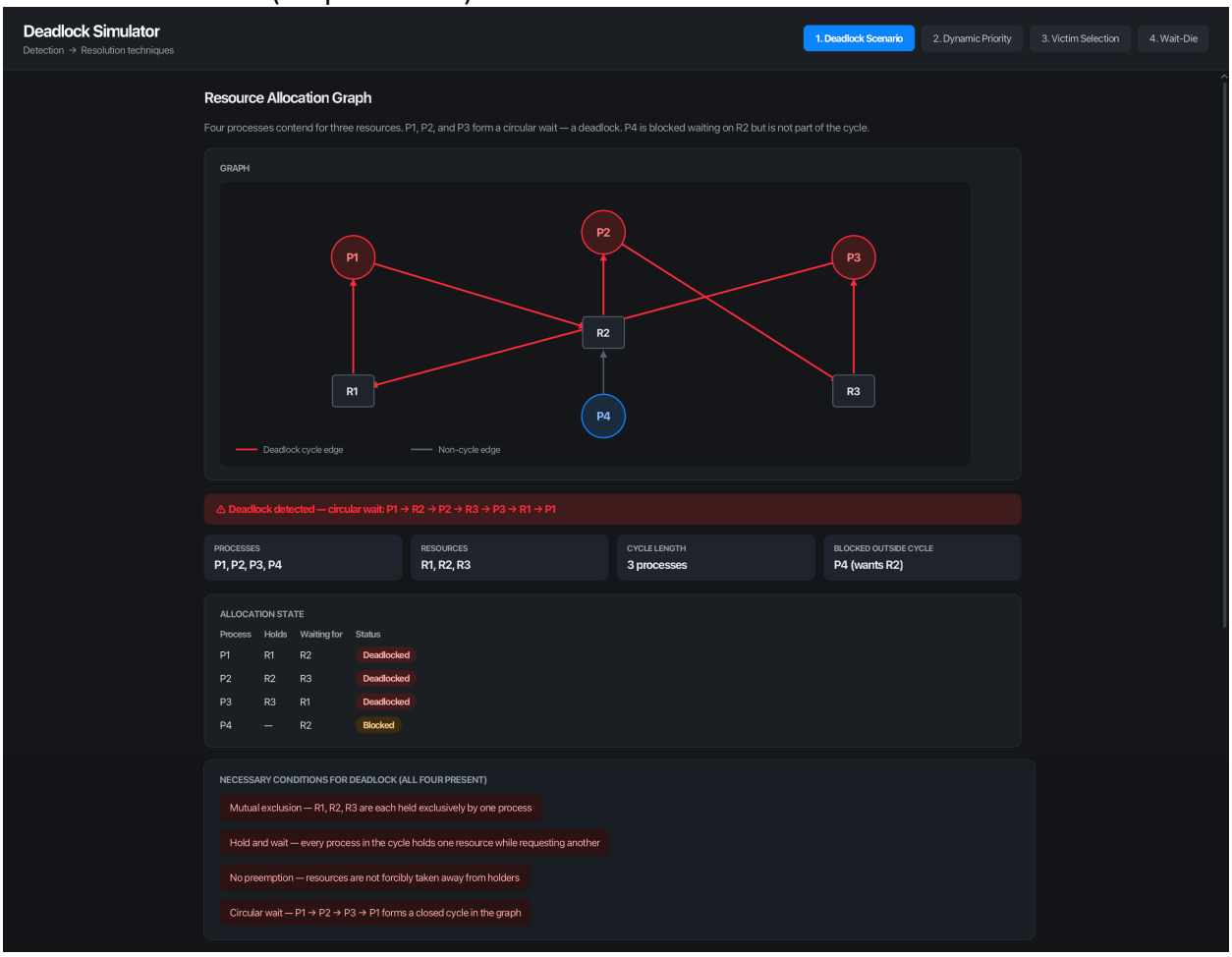
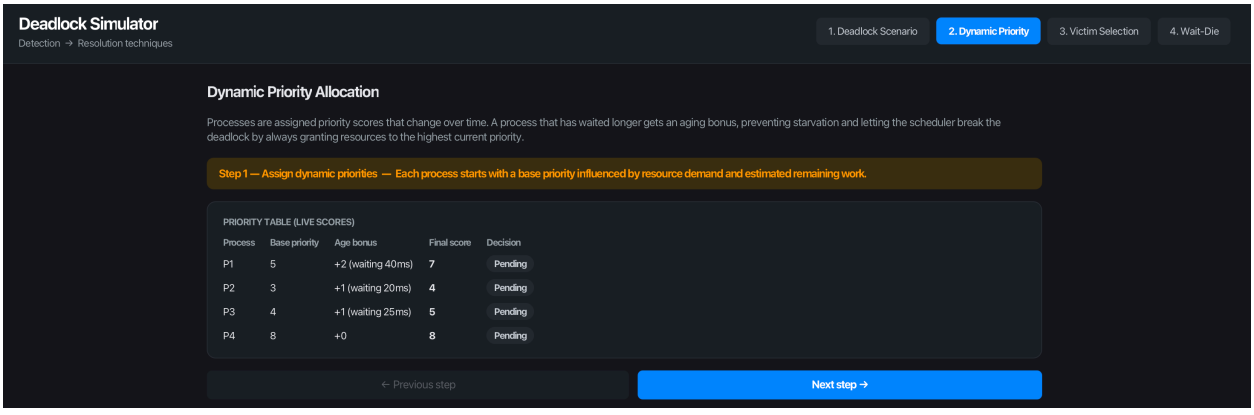
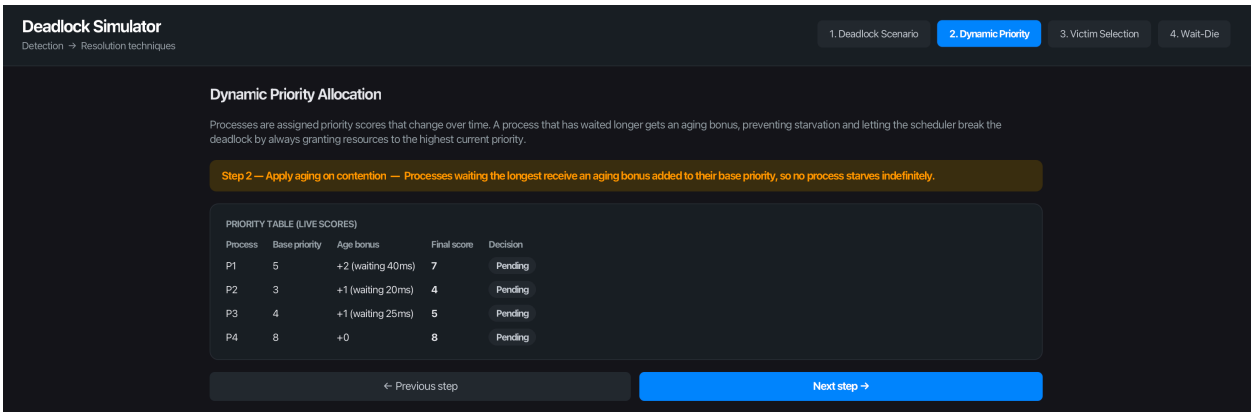
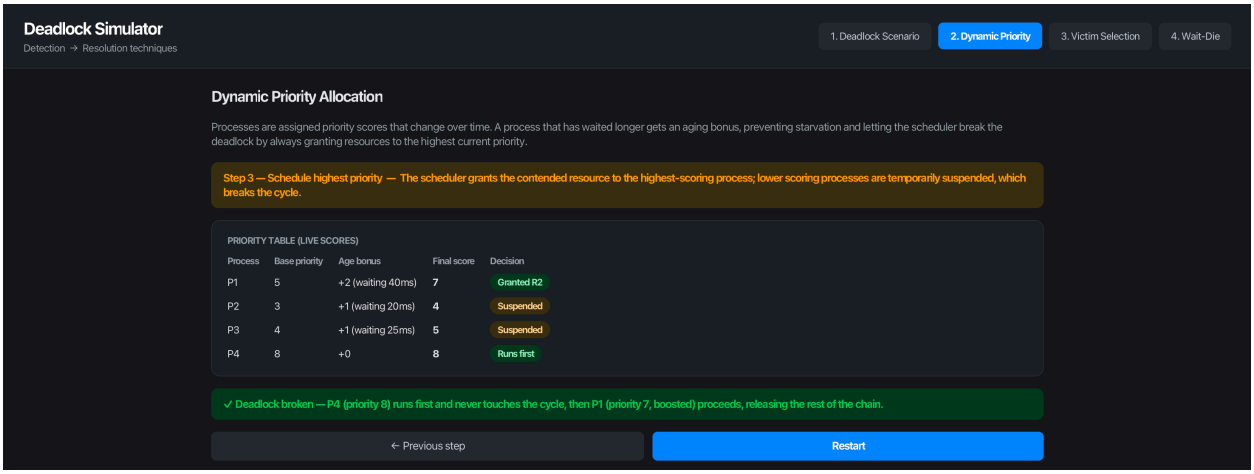
Deadlock Simulation (No prevention)



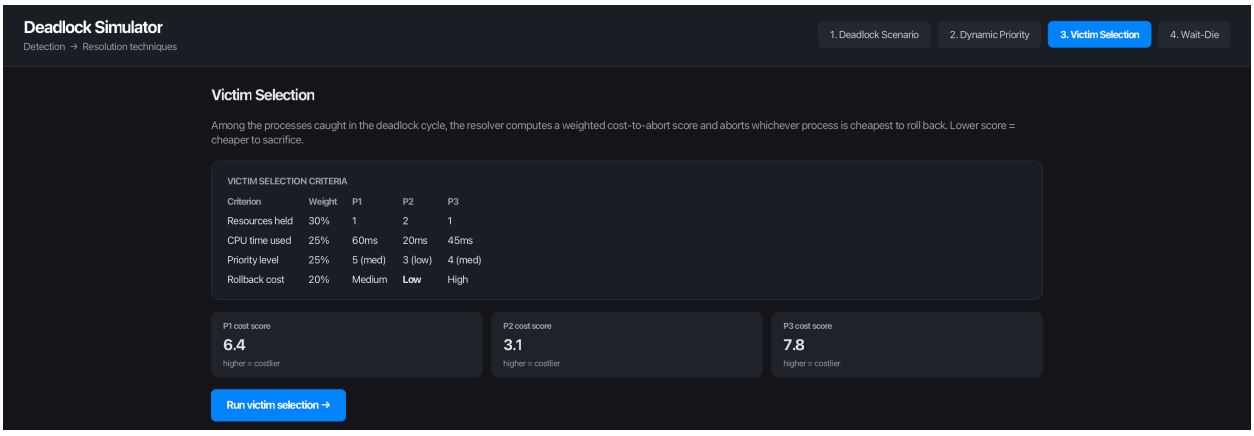
Prevention Technique 1: Dynamic Priority Allocation

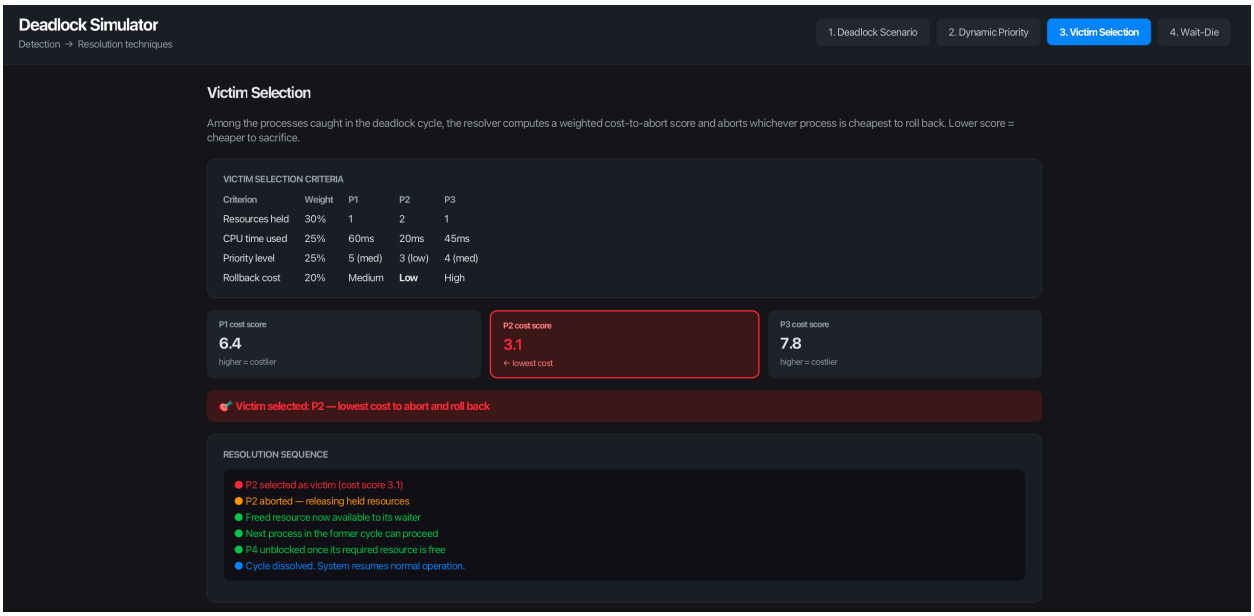




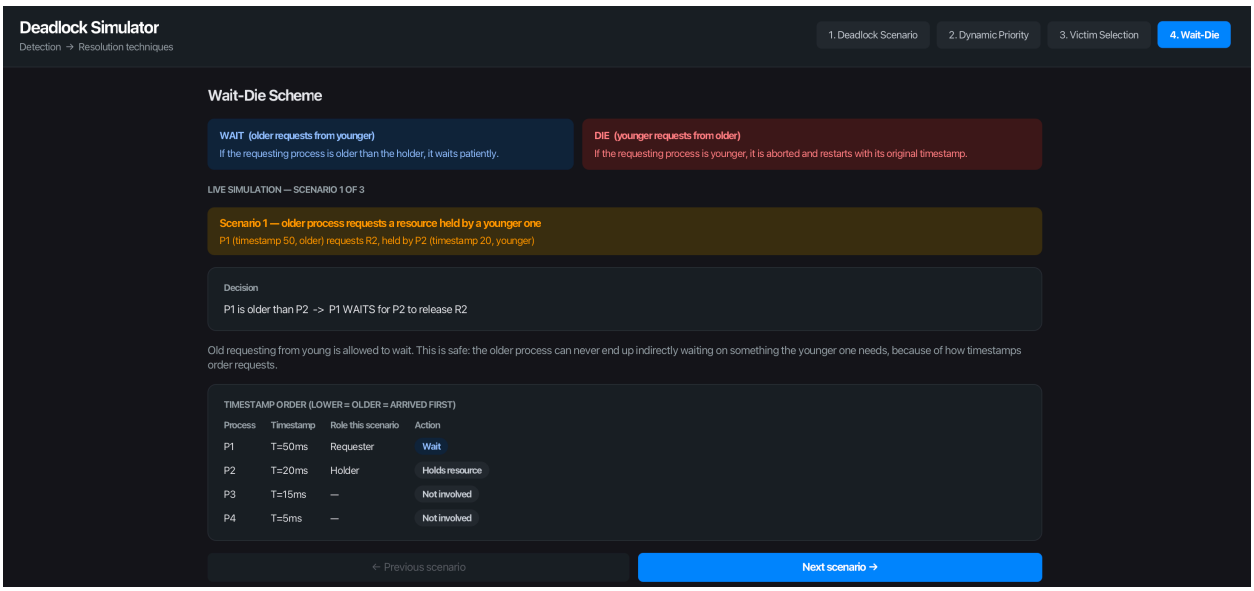


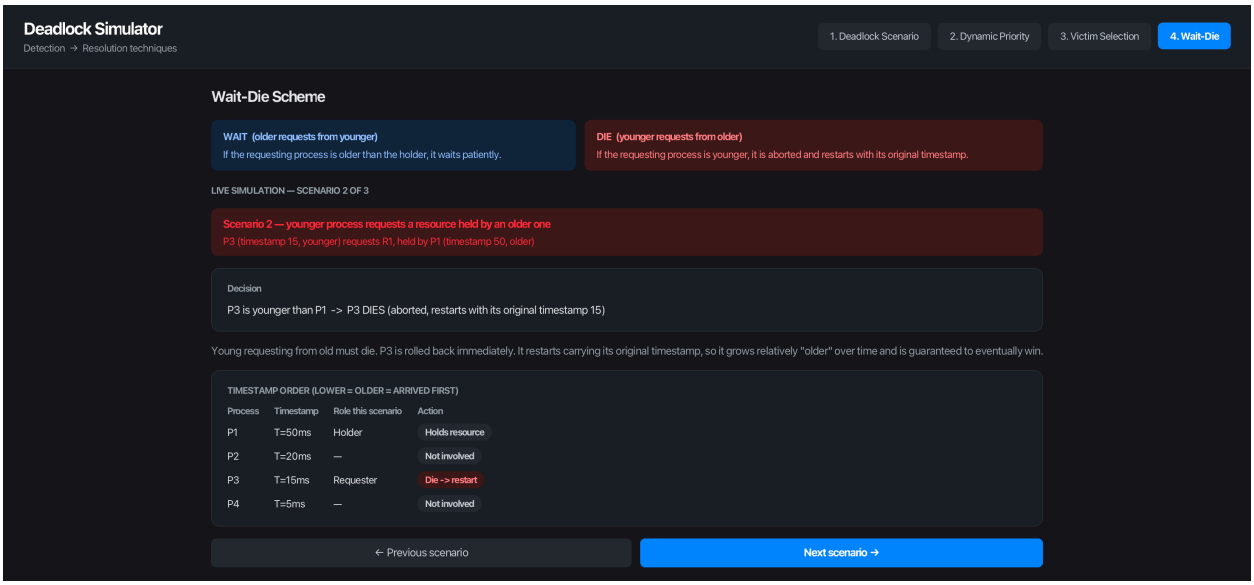
Prevention Technique 2: Victim Selection

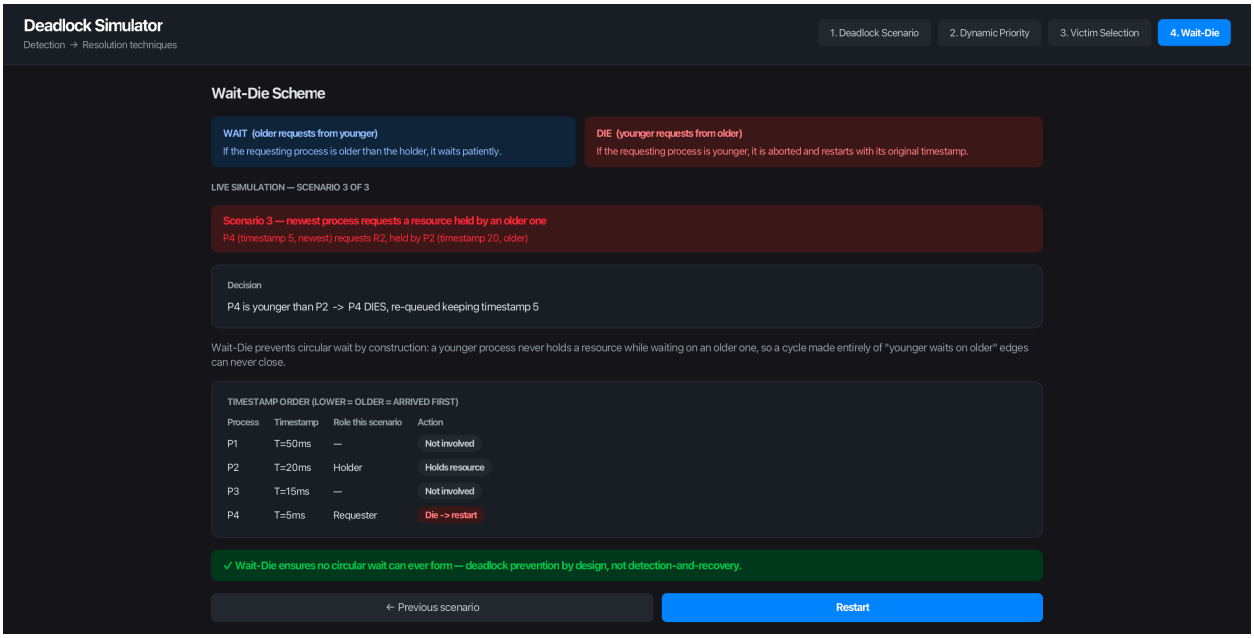




Prevention Technique 3: Wait-Die




Deadlock Simulator
Detection → Resolution techniques
1. Deadlock Scenario
2. Dynamic Priority
3. Victim Selection
4. Wait-Die
Wait-Die Scheme
WAIT (older requests from younger)
If the requesting process is older than the holder, it waits patiently.
DIE (younger requests from older)
If the requesting process is younger, it is aborted and restarts with its original timestamp.
LIVE SIMULATION — SCENARIO 2 OF 3
Scenario 2 — younger process requests a resource held by an older one
P3 (timestamp 15, younger) requests R1, held by P1 (timestamp 50, older)
Decision
P3 is younger than P1 -> P3 DIES (aborted, restarts with its original timestamp 15)
Young requesting from old must die. P3 is rolled back immediately. It restarts carrying its original timestamp, so it grows relatively "older" over time and is guaranteed to eventually win.
TIMESTAMP ORDER (LOWER = OLDER = ARRIVED FIRST)
Process Timestamp Role this scenario Action
P1 T=50ms Holder Holds resource
P2 T=20ms — Not involved
P3 T=15ms Requester Die -> restart
P4 T=5ms — Not involved
← Previous scenario
Next scenario →


Deadlock Simulator
Detection → Resolution techniques
1. Deadlock Scenario
2. Dynamic Priority
3. Victim Selection
4. Wait-Die
Wait-Die Scheme
WAIT (older requests from younger)
If the requesting process is older than the holder, it waits patiently.
DIE (younger requests from older)
If the requesting process is younger, it is aborted and restarts with its original timestamp.
LIVE SIMULATION — SCENARIO 3 OF 3
Scenario 3 — newest process requests a resource held by an older one
P4 (timestamp 5, newest) requests R2, held by P2 (timestamp 20, older)
Decision
P4 is younger than P2 -> P4 DIES, re-queued keeping timestamp 5
Wait-Die prevents circular wait by construction: a younger process never holds a resource while waiting on an older one, so a cycle made entirely of "younger waits on older" edges can never close.
TIMESTAMP ORDER (LOWER = OLDER = ARRIVED FIRST)
Process Timestamp Role this scenario Action
P1 T=50ms — Not involved
P2 T=20ms Holder Holds resource
P3 T=15ms — Not involved
P4 T=5ms Requester Die -> restart
✓ Wait-Die ensures no circular wait can ever form — deadlock prevention by design, not detection-and-recovery.
← Previous scenario
Restart